Ji 2020 cutaneous SCC — analysis summary

RUN 5 OF 10

GO / NO-GO GATES

Gate 1 — integration & continuity → GO

Harmony (θ=5) merges 8 samples into one connected kNN component; continuous, no sample islands.

Gate 2 — module geography → GO (branching)

Gast/pSom-high hub diverges into a Gast-arm and a pSom-arm; Crest a terminal readout toward the tips.

Gate 3 — hub cross-repression → GO (branching)

Hub Gast↔pSom module r = −0.37 (perm p = 0.007), competitive; Puram −0.49 / Filbin −0.34 anchors.

STEP FIGURES

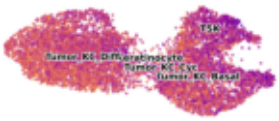
Manifold — samples



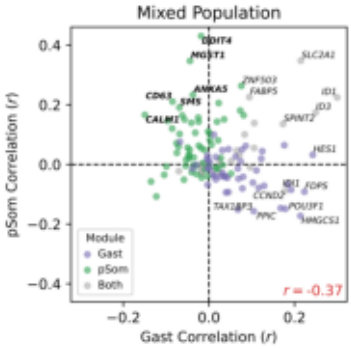
Trajectory / root



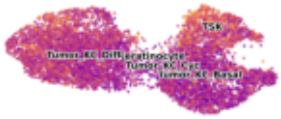
Gastrulation



Gate 3: r = −0.37



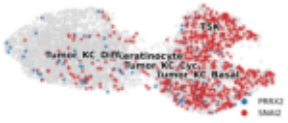
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
Cutaneous SCC (Ji et al., 2020), 10x — 10,531 malignant cells over 8 samples after thresholding (8 of 10; clean 6.7× per-sample count gap, keep ≥100). Developmental compartment 6,800 cells (3,731 cycling scoped out).

Verdict.
The embryonic pattern reproduces as a BRANCHING hub-and-spoke — a Gast/pSom-high hub diverging into competing Gast and pSom arms with hub cross-repression r = −0.37, the Puram signature on squamous biology. Gate 1 GO / Gate 2 GO / Gate 3 GO. EMT-TFs SNAI2 (50%, late/terminal) + PRRX2 (9%, root-end).

Parameter choices

Min-cell threshold	≥100 per sample (6.7× count gap 40→268); drops P3, P5	**Overdispersed genes**	1,443 (scFates find_overdispersed)
Embedding	PCA 50 → Harmony(sample, θ=5) → kNN n_pcs=20 (scree elbow PC17), UMAP spread=1.3	**Trajectory / geometry**	scFates PPT tree (branching), Nodes=50 λ=2000 σ=0.5; dev compartment, fold-free
Root (hub)	Gast/pSom-high, Crest-low convergence node (Crest late = terminal)	**Hub identity**	Hub milestone, 1,195 cells; module mutual r = −0.37 (p=0.007)
Crest arm	terminal readout toward arm tips (ρ_pt = +0.25); not on the cross-repression axis	**EMT-TFs**	SNAI2 (50%, late/terminal) + PRRX2 (9%, root-end); SNAI1/ZEB1 near 10x floor